MILITARY SPECIFICATION SHEET

ACTUATOR, ROTARY, HYDRAULIC 2-POSITION, RACK-AND-PINION TYPE 5,000 TO 92,000 INCH-POUNDS TORQUE 4-BOLT MOUNTING

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-A-24533.

GENERAL

Part number. Rotary actuators shall be designated by a part or identifying number (PIN). The PIN shall consist of the following:

M	24533	/2	-	WW	X	Y	Z
Prefix to indicate item defined by inch-pound units	Specification number	Applicable specification sheet number		Torque	Rotation angle	Dashpot designation	Environmental grade

Applicable specification sheet number. The applicable specification sheet number designates the type of actuator, number of positions, torque range and type of mounting. The mounting configuration shall be as specified on figure 1.

Torque. A two digit code shall classify the torque output of the rotary actuator as specified in table I.

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TABLE I Torque ranges.

Code	Torque (rein) at 2000 lb/in² (in-lb)	Torque (max) at 3000 lb/in² (in-lb)
05	3,440	5,120
13	7,300	12,700
17	9,300	16,600
28	16,000	27,500
43	24,000	43,000
92	53,456	91,560

Rotation angle. A single letter shall describe the rotation angle of the rotary actuator as specified in table II.

TABLE II. Rotation angle.

Code	Rotation angle (degrees)
Q	90
H	180
X	As identified

Dashpot. The dashpot for the rotary actuator shall be designated by a single digit as follows:

Code	Description
1	Actuator with no dashpots.
2	Actuator with dashpots in both clockwise and counterclockwise directions.
3	Actuator with dashpot in clockwise direction only.
4	Actuator with dashpot in counterclockwise direction only,

Environmental grade. The actuator environmental grade shall be designated by a single letter as follows:

Code	Description
A	Standard actuator.
B	Actuator for extremely corrosive environment.

The requirements for each environmental grade are defined in MIL-A-24533,

Construction.

Envelope dimensions. Rotary actuators envelope dimensions shall be as specified on figure 1.

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Mounting. The location and size of the mounting holes shall be as specified on figure 1.

Shaft. The actuator shall be furnished with an internal spline shaft as specified on figure 2. The nominal diameter of the shaft shall be as specified on figure 1.

Ports. The actuator shall be furnished with ports in accordance with figure 3, The size and location of the ports shall be as specified on figure 1.

DESIGN

Operating pressure. The rotary actuator shall be designed for 3,000 pounds per square inch (lb/in^2) operating pressure.

Torque output. The rotary actuator shall provide the torque output specified in table I with the specified differential pressure across the actuator.

Rotational tolerance. The rotational tolerance for the actuator shall not exceed plus 2 degrees to minus 0 degree for full rotation.

Internal leakage. Internal leakage from the pressurized piston chambers into the gear case shall not exceed 0.1 cubic centimeter in 5 minutes.

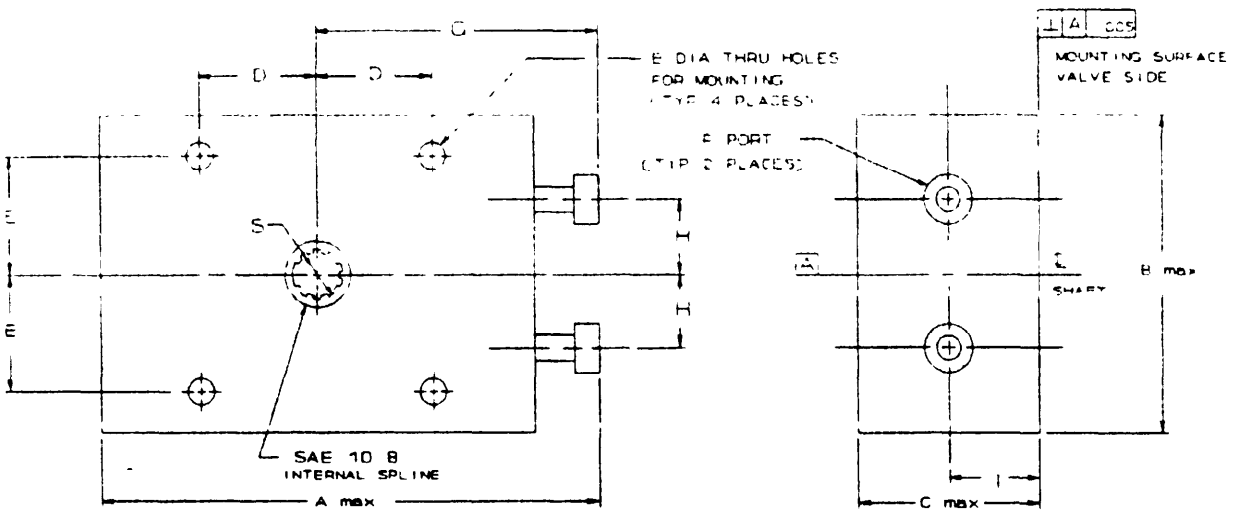
Gear case lubricating fluid. The gear case shall be filled to the three-quarter level with MIL-L-17331, 2190 TEP fluid.

Operating cycles. Actuators shall have a minimum operating life of 50,000 cycles, Packings or seals shall operate for 15,000 cycles before replacement is required.

Relief ports. A relief port and a drain relief valve for the gear chamber of the actuator shall be provided for all actuators at a location which is accessible without disassembly of the actuator. The gear chamber shall be designed for 200 $lb\backslash in^2$ and shall withstand 300 lb/in^2 proof pressure without damage or leakage. The relief valve shall be set to open at a nominal gauge pressure of 200 lb/in^2.

Preparing activity;
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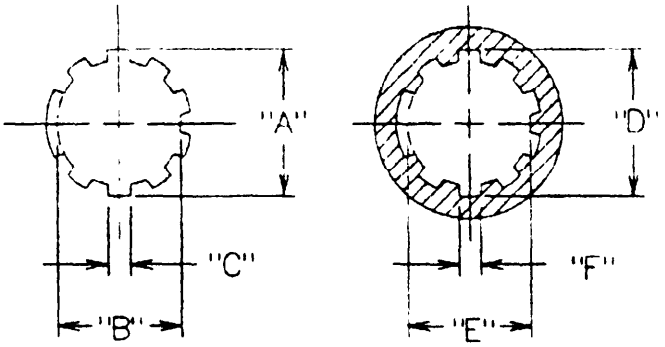
TORQUE	TORQUE PART NUMBER DESIGNATION	NOMINAL SHAFT DIAMETER S	ROTATING ANGLE (DEGREES)	MAXIMUM ENVELOPE			MOUNTING			PORTS			
				A	B	C	D	E	F	G	H	I	P*
500	005	1.375	090	11.50	6.75	3.50	1.13	1.75	0.781	6.00	1.25	2.00	B
			180	13.50						7.00			
			183	13.75						7.13			
13000	013	1.375	090	13.50	7.33	6.00	1.687	3.375	0.375	7.00	1.44	2.88	B
			180	15.90						8.21			
			183	16.13						8.32			
17000	017	1.500	090	13.60	7.62	3.81	2.88	2.88	0.656	7.00	1.81	2.12	B
			180	17.00						8.70			
28000	028	2.000	090	17.00	13.88	6.94	3.50	5.88	1.063	9.62	2.50	1.88	B
			180	22.00						12.12			
43000	043	2.250	090	18.09	9.50	6.00	3.00	4.00	0.656	9.30	2.25	2.75	B
			180	25.00						12.94			
93000	092	3.500	090	17.87	12.50	7.19	3.25	4.25	0.781	9.13	3.25	3.31	D
			180	23.00						11.75			

NOTE: All dimensions are in inches.
* Refer to port detail in MIL-A-24533.

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FIGURE 1. Rotary actuator envelope configuration.

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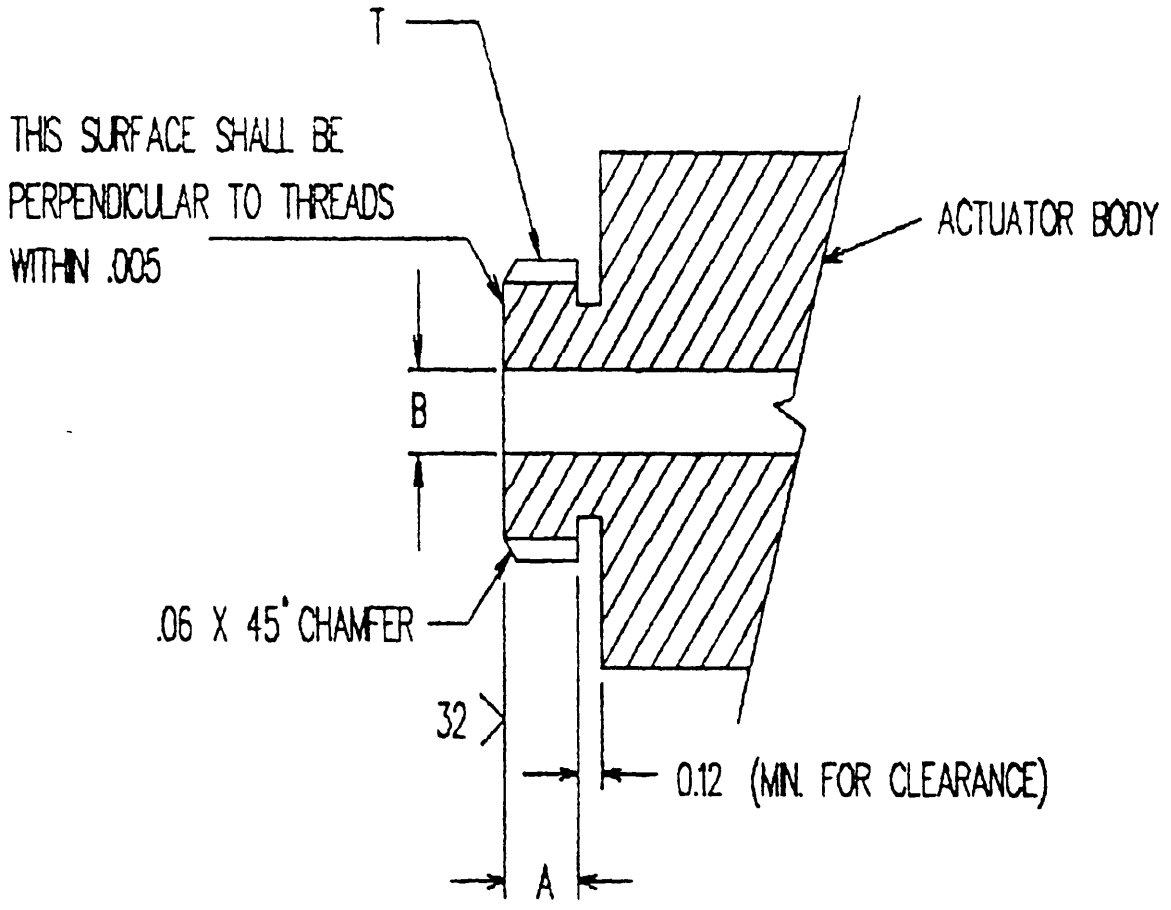


NOMINAL O.D.		SHAFT			FITTING		
HOB O.D FOR DESIGN RUNOUT	SHAFT & FITTING	"A" O.D.	"B" RT DIA	"C" WIDTH	"D" O.D.	"E" RT DIA	"F" WIDTH
2.000" MIN	0.500"	0.498 0.497	0.424 0.419	0.075 0.074	0.502 0.501	0 430 0.429	0.078 0.076
	0 625"	0.623 0.622	0.532 0.527	0.095 0.094	0.627 0.626	0.538 0.537	0.098 0.096
	0.750"	0.748 0.747	0.639 0.634	0.114 0.113	0.752 0.751	0.645 0 644	0.117 0.115
	0.875"	0.873 0.872	0.747 0.742	0.134 0.133	0.877 0.876	0 753 0 752	0.137 0.135
	1 000"	0.998 0 997	0.854 0 849	0 153 0.152	1.002 1 001	0.860 0.859	0.156 0.154
	1.125"	1 123 1 121	0 962 0 957	0.173 0.172	1.127 1 126	0.968 0.967	0 176 0.174
	1.250"	1.248 1.246	1 069 1.064	0.192 0 191	1.252 1.251	1.075 1.074	0.195 0.193
	1.375"	1.373 1.371	1.177 1.172	0.212 0.211	1.377 1.376	1.183 1.182	0.215 0 213
	1 500"	1.498 1 496	1.284 1.279	0.231 0.230	1 502 1.501	1.290 1.289	0.234 0.232
2.125" MIN	1.625"	1.623 1 621	1.392 1 387	0.251 0.250	1.627 1.626	1.398 1.397	0.254 0.252
	1.750"	1 748 1 746	1.499 1.494	0.270 0.269	1.752 1 751	1.505 1.504	0.273 0.271
	2.000"	1.997 1.995	1 713 1.708	0.308 0.307	2 002 2.000	1.720 1 718	0.312 0.309
2 250" MIN	2.250"	2.247 2.245	1.928 1.923	0.347 0.346	2.252 2 250	1.935 1 933	0.351 0.348
	2 500"	2.497 2.495	2 143 2 138	0.386 0 385	2.502 2 500	2.150 2.148	0.390 0.387
	3 000"	2.997 2 995	2 573 2 568	0 464 0 463	3 002 3 000	2.580 2 578	0 468 0 465

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FIGURE 2. Spline configuration.

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P PORT	A	B	T
A	0.375	0.219	1.000 - 14UNS-3A
B	0.500	0.313	1.188 - 12UN-3A
C	0.500	0.438	1.375 - 12UNF-3A
D	0.500	0.500	1.750 - 12UN-3A

All dimensions are in inches.

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FIGURE 3. Port configuration.

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